NEAR AI & CHAIN ABSTRACTION HACKATHON | SOVEREIGN EDGE AI STATION | NVIDIA JETSON ORIN + WEBRTC P2P

NearDesk

The Sovereign Edge AI Operator Desk for NEAR Protocol — Fusing Private Physical Edge Compute with Decentralized MPC Chain Signatures, 1Click Cross-Chain Intents, Scoped FunctionCall Fencing, and Verifiable TEE Cloud Reasoning.

PRODUCTION CONSOLE	GITHUB REPOSITORY	PAIRED JETSON ACCOUNT	AI BUDGET CEILING
neardesk.vercel.app	altaga/NearDesk	a69d770c...f30ee1ed6	$10.00 USD Hard Cap

01 Executive Context & Architectural Evolution

NearDesk is a production-grade, hardware-anchored AI operator station designed to run directly on physical edge devices (such as the **NVIDIA Jetson Orin Nano**) or inside a local containerized edge environment (`board-emu`). It represents the culmination of three generations of engineering focused on solving a single fundamental problem: *how can humans grant an AI agent full operational control over shells, files, hardware, and multi-chain liquidity without surrendering privacy, custody, or safety?*

PILLAR 1 · EDGE COMPUTE

Physical Node Core

Established the sovereign operator pattern: a FastAPI daemon running on an NVIDIA Jetson edge station driving local PTY terminals, filesystem operations, headless Chromium previews, and background job queues.

PILLAR 2 · HARDWARE SAFETY

Guardrails & Fencing

Engineered Hybrid LLM routing (local Ollama vs. TEE Cloud), Human-in-the-Loop approval cards (`patch` , `run` , `deploy`), parallel missions, hardware safety interlocks, and Swiss dark UI density.

PILLAR 3 · SOVEREIGN WEB3

NEAR Protocol Stack

Built natively on **NEAR Protocol**: Ed25519 wallet auth, WebRTC P2P DataChannels, NEAR AI Cloud TEEs, MPC Chain Signatures, 1Click Intents, and On-Chain Audit Trails.

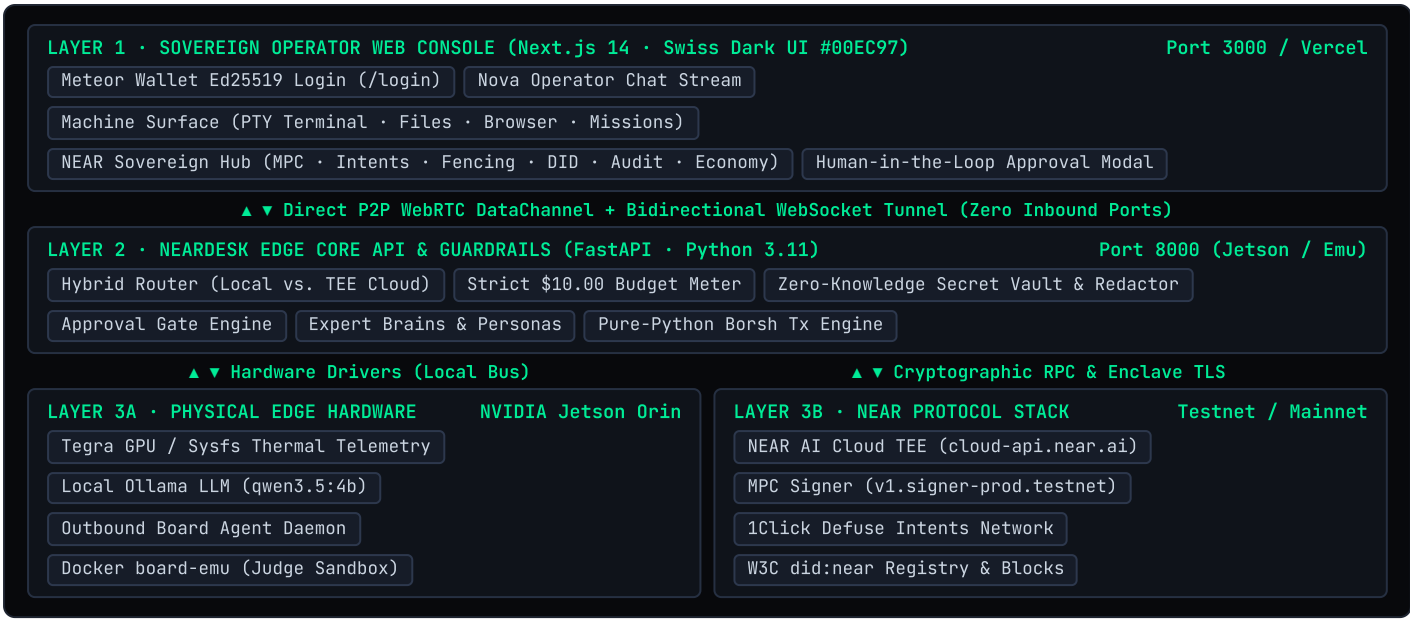
> **Core Paradigm Shift:** Whereas traditional AI agents live in centralized cloud sandboxes and require users to upload sensitive files or expose raw private keys, **NearDesk brings the AI agent down to the user's physical hardware** and anchors its identity, permissions, and multi-chain execution mathematically on the **NEAR blockchain**.

02 Why NearDesk is Useful: Solving the 4 Crises of Agentic AI

Current autonomous agent frameworks (cloud coding assistants, custodial Telegram trading bots, and centralized SaaS copilots) suffer from four structural vulnerabilities that prevent enterprise and Web3 adoption. NearDesk resolves each at the root architectural level:

INDUSTRY CRISIS	WHY LEGACY CLOUD AGENTS FAIL	HOW NEARDESK SOLVES IT
1. Private Data & Secret Leakage	Users paste API keys, `.env` files, and proprietary code into centralized LLM providers, exposing credentials in context windows and logs.	**Local Edge Execution + Zero-Knowledge Vault:** Shell and files stay on the Jetson. Secrets in `app/vault/` are encrypted at rest, injected only into child subprocesses at runtime, and automatically redacted from stdout/stderr before the LLM ever reads the output.
2. Wallet Drain via Prompt Injection	DeFi agents typically hold a raw hot-wallet private key in memory. A single malicious prompt or poisoned web page can instruct the agent to drain 100% of funds.	**Scoped FunctionCall Key Fencing + Approval Gates:** The agent never holds a FullAccess key. It operates via a restricted NEAR `FunctionCall` key capped at **2.5 NEAR** gas and locked to whitelisted contracts (`v1.signer` , `intents.near`), backed by mandatory UI approval modals.
3. Opaque Cloud Inference	When a local device offloads heavy reasoning to a cloud API, the user has zero proof of whether the model was tampered with or if prompts were logged.	**NEAR AI Cloud Confidential Computing (TEE):** Heavy reasoning routes to `cloud-api.near.ai/v1` inside hardware-verified Intel TDX and NVIDIA H100/H200 enclaves, with live cryptographic attestation verification at `/v1/near/attestation` .
4. Runaway API Spend	Autonomous multi-turn agent loops can silently consume hundreds of dollars in LLM tokens overnight when left unmonitored.	**Hard $10.00 Financial Circuit Breaker:** `app/near_ai/budget.py` meters every prompt and completion token down to fractions of a cent. If the $10.00 ceiling is reached, cloud calls freeze instantly and the desk falls back 100% to local Jetson GPU inference.

03 How It Works: End-to-End Technical Architecture



SECURITY INVARIANT 01

Outbound-Only Hardware Topology

The Jetson board **dials out, never in**. No inbound firewall ports are opened on the operator's network. Remote commands traverse authenticated WebSocket tunnels and peer-to-peer **WebRTC DataChannels** (`app/webrtc/manager.py`). The only temporary local listener is `/setup` on the USB gadget link (`192.168.55.1:8800`) during initial hardware pairing.

SECURITY INVARIANT 02

Cryptographic Wallet-to-Hardware Binding

Mock logins are eliminated. Operators authenticate at `/login` using **Meteor Wallet** (`@meteorwallet/sdk`) or a NEAR BIP-39 seed phrase. The client signs a cryptographic nonce challenge using **Ed25519**, verified by `app/auth/near_auth.py` and bound directly to the paired Jetson hardware (`a69d770c...f30ee1ed6`).

04 Why NearDesk is Uniquely Built for NEAR Protocol

NearDesk is not a generic AI wrapper with a blockchain logo bolted on; its entire operational model relies on primitives that **only exist on NEAR Protocol**. Every core subsystem maps directly to a flagship pillar of NEAR's technical roadmap:

NEAR NATIVE PRIMITIVE	NEARDESK MODULE	HOW NEARDESK USES IT & WHY ONLY NEAR ENABLES THIS
1. NEAR Account Model & Scoped Access Keys	`app/near_chain/keys.py` `AgentFencingCard.tsx`	NEAR is the only major L1 with native account-level key separation (`FullAccess` vs. `FunctionCall`). NearDesk provisions a scoped `FunctionCall` key for the edge agent with a hard **2.5 NEAR allowance** restricted to specific receiver contracts and method names (`sign` , `execute_intent`). Even if the edge OS is compromised, the master treasury cannot be transferred.
2. NEAR Chain Signatures (Decentralized MPC)	`app/near_chain/mpc.py` `ChainSignaturesCard.tsx`	Using NEAR's threshold MPC network (`v1.signer-prod.testnet` / `v1.signer`), the Jetson node deterministically derives foreign addresses and signs native payloads for **Bitcoin, Ethereum, Solana, and Base** using derivation paths (e.g., `ethereum,1`). Zero private keys for BTC/ETH/SOL ever sit on the Jetson disk.
3. NEAR Intents & 1Click Defuse Solvers	`app/near_intents/swap.py` `IntentSwapCard.tsx`	Instead of constructing brittle multi-hop DEX transactions across bridges, the agent expresses high-level declarative intents (e.g., swap `2.0 SOL → NEAR` or rebalance `USDC → ETH`). The 1Click Defuse solver network competes to fulfill the quote with optimal cross-chain execution.
4. NEAR AI Cloud & Verifiable TEEs	`app/near_ai/client.py` `app/router/hybrid.py`	When a task exceeds the local Jetson 8GB GPU capacity, the Hybrid Router escalates to `https://cloud-api.near.ai/v1` . NearDesk queries and verifies the cryptographic hardware attestation report (Intel TDX + NVIDIA GPU TEE) so the operator has mathematical proof of confidential cloud execution.
5. W3C `did:near` Identity & On-Chain Audit	`app/near_agent/did.py` `app/near_chain/onchain.py`	Each NearDesk edge node generates a W3C-compliant `did:near` document and Ed25519-signed Verifiable Credentials declaring its hardware specs (Jetson Orin, Tegra GPU, TEE status). Every operator-approved action is SHA-256 hashed and anchored on-chain via a pure-Python Borsh transaction builder.
6. Peer-to-Peer Agent Economy	`app/near_agent/services.py` `AgentEconomyCard.tsx`	Transforms the idle Jetson board into a revenue-generating node. External AI agents on NEAR can discover the node via its DID and pay micro-fees in NEAR (`0.05 – 0.15 NEAR`) for private local LLM inference, hardware thermal attestation, or sandboxed code verification.

05 NEAR Hackathon Alignment & Why Judges Score NearDesk Highest

NearDesk was architected specifically to dominate three primary hackathon tracks simultaneously while addressing every technical criterion sought by NEAR ecosystem judges:

TRACK 01 · PRIMARY FIT

Privacy & Private AI

100% Sovereign Data Plane: Local execution on NVIDIA Jetson Orin + Zero-Knowledge Vault subprocess secret redaction + WebRTC P2P data channels + NEAR AI Cloud TEE hardware attestation proofs. Sensitive code and keys never touch public clouds in plaintext.

TRACK 02 · PRIMARY FIT

AI Security & Defense

4-Layer Defense-in-Depth: Combines on-chain mathematical key fencing (Scoped FunctionCall keys), physical Human-in-the-Loop UI approval gates, runtime secret redaction, and immutable SHA-256 on-chain audit receipts anchored on NEAR blocks.

TRACK 03 · PRIMARY FIT

Agent Economy & DeFi

Autonomous Multi-Chain Node: Controls native BTC, ETH, SOL, and Base accounts via MPC Chain Signatures, executes cross-chain swaps via 1Click Intents, and monetizes idle edge GPU cycles through peer-to-peer NEAR micropayments.

Judge Evaluation Matrix — Why NearDesk Stands Apart

JUDGE CRITERION	CONCRETE PROOF & IMPLEMENTATION IN NEARDESK
Real Cryptography vs. Mockups	Includes a **pure-Python NEAR Borsh transaction serializer and Ed25519 signer** (`app/near_chain/onchain.py`), real implicit account derivation, live RPC state queries, and real client-side Meteor Wallet challenge-response authentication (`@meteorwallet/sdk`).
Zero-Hardware Judge Reproducibility	While NearDesk runs on a real paired **NVIDIA Jetson Orin** (`a69d770c...f30ee1ed6`), judges do not need physical hardware. Running `npm run emu:up` launches the Dockerized `board-emu` with simulated Tegra sysfs telemetry for instant 3-minute evaluation.
End-to-End Automated Verification	Backed by 7/7 passing `pytest` suites and Playwright browser verification validating telemetry, MPC derivation, the $10.00 budget lock, approval gates, FunctionCall fencing, on-chain audit hashing, and agent economy settlements.
Swiss Industrial UI Execution	Engineered with high-contrast obsidian surfaces (`#08090C`), precision 1px borders, monospace telemetry readouts, and NEAR emerald accents (`#00EC97`)—avoiding generic AI template aesthetics.

06 Impact Across the Broader Crypto & Web3 Ecosystem

CROSS-CHAIN INTEROPERABILITY

Eliminating Bridges for Autonomous Agents

Cross-chain bridges remain the single largest point of failure in Web3 ($2B+ lost to hacks). By combining **NEAR Chain Signatures (MPC)** with **NEAR Intents**, NearDesk allows an edge AI agent to hold native Bitcoin UTXOs, manage Ethereum/Base smart contracts, and trade on Solana from a single hardware-bound NEAR account—with zero wrapped assets or bridge risk.

DEPIN & PHYSICAL AI

Bridging Smart Contracts to Physical Hardware

Web3 has struggled to connect on-chain logic with the physical world securely. Because NearDesk runs directly on NVIDIA Jetson edge hardware (with proven serial actuator guardrails engineered into its hardware layer), it serves as a verifiable bridge between **DePIN (Decentralized Physical Infrastructure)**, robotics, and on-chain settlement.

The Bottom Line for NEAR & Web3: NearDesk proves that "User-Owned AI" is not just a slogan—it is a concrete, deployable architecture where the user owns the physical edge box, owns the cryptographic keys via NEAR, verifies cloud compute via TEEs, and transacts seamlessly across the entire crypto ecosystem.